

Melting Ice Trackers

Complete Project Pipeline & Development Roadmap

Core idea Use repeated satellite observations to identify glacier ice, measure ice-covered area, detect spatial and temporal change, and present notable changes for human investigation.	Project principle Observe → Identify → Measure → Compare → Detect → Visualize → Investigate
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THE END-TO-END PIPELINE

1	Select Glaciers Choose the real-world glacier(s) to monitor.
2	Collect Satellite Imagery Acquire repeated, georeferenced observations over time.
3	Preprocess Images Prepare bands, clouds, alignment, cropping and normalization.
4	Create / Obtain Ice Masks Define which pixels represent the target glacier ice.
5	Train Segmentation Model Teach an AI model to predict ice at pixel level.
6	Predict Ice Masks Run the trained model on new satellite observations.
7	Calculate Ice Area Convert segmented pixels into real-world geospatial area.
8	Store Historical Observations Build a time series of measurements for each glacier.
9	Change Detection Quantify absolute and percentage changes.
10	Spatial Change Analysis Locate where glacier extent has changed.

11	Significance / Anomaly Analysis Compare observed change with historical behavior.
12	Dashboard & Map Present measurements, maps, imagery and history.
13	Reports / Alerts Generate traceable summaries and investigation flags.

1. UNDERSTAND THE PROJECT OBJECT

Glacier = a persistent body of land-based ice formed from accumulated snow that flows slowly under its own weight. For this project, the glacier is the real-world object being monitored.

One glacier is NOT one image. A glacier should have multiple satellite observations at different dates so the system can study change over time.

Real-world object	Observation	AI output	Analytics
Glacier G001	Satellite image — 2024	Ice mask	Ice area / change
Same glacier	Satellite image — 2025	Ice mask	Comparison

Scientific boundary

- Observed decrease in ice-covered area is not automatically the same as total ice volume melted.
- Snow, seasonal conditions, clouds, shadows and image quality can affect what the satellite sees.
- The system should report observed satellite-derived changes and uncertainty rather than claim a specific cause without supporting evidence.
- A notable-change flag means 'worth investigating', not 'a disaster has been proven'.

2. PHASE 1 — DATA FOUNDATION

Select glacier(s) → choose imagery source → collect repeated observations → inspect and catalogue them.

- Start with one real-world glacier so the complete pipeline can be validated end-to-end.
- Collect multiple dates of the same glacier and document seasonal conditions.
- Record date, sensor, product ID, location/bounds, CRS, spatial resolution, cloud information and file path.
- Separate raw imagery from processed data and maintain metadata for traceability.

3. PHASE 2 — IMAGE PREPROCESSING

Raw satellite scenes are converted into consistent inputs for the AI pipeline.

- Handle or mask cloud-contaminated regions when appropriate.
- Choose the spectral bands/features needed by the segmentation approach.
- Ensure image layers and observations are spatially consistent.
- Crop or tile large scenes to the glacier region / training regions.
- Apply a consistent numerical preprocessing strategy.

4. PHASE 3 — ICE MASK DATASET

Create ground-truth masks that teach the model which pixels belong to glacier ice.

- Satellite image = model input.
- Mask = pixel-level label map; conceptually 1 = target glacier ice and 0 = non-target.
- Every labeled sample connects the same geographic pixels in image and mask.
- Split data into training, validation and test sets.

5. PHASE 4 — AI SEGMENTATION

Train a semantic-segmentation model to predict glacier-ice pixels.

- Input: satellite image or selected spectral-band stack.
- Possible models: U-Net, DeepLab, SegFormer or another suitable architecture.
- Training compares predicted masks with ground truth and updates model weights.
- Evaluate using IoU, Dice and visual inspection of failure cases.

6. IMAGE → MASK → AREA

This is the central AI-to-measurement transition.

Satellite observation	Predicted mask	Geospatial measurement
Pixels containing mountains, rock, snow, glacier ice, water, cloud, etc.	Pixel-level classification identifying the target glacier-ice region.	Use raster geospatial information to calculate physical area represented by selected pixels.

Conceptual numerical example

If a raster has 10 m × 10 m ground sampling and 10,000 pixels are classified as target ice, the simple conceptual calculation is $10,000 \times 100 \text{ m}^2 = 1,000,000 \text{ m}^2 = 1 \text{ km}^2$. In the actual system, the calculation must respect the raster's georeferencing, CRS/projection and pixel geometry.

The AI mask is not the final scientific measurement. Geospatial processing converts the mask into a physically meaningful area measurement.

7. HISTORICAL MONITORING & CHANGE ANALYSIS

Once every observation has an ice-area measurement, the project becomes a monitoring system rather than a one-image classifier.

Date	Ice area	Change	Change %
2020	62.4 km ²	—	—
2021	61.8 km ²	−0.6 km ²	−0.96%
2022	60.9 km ²	−0.9 km ²	−1.46%
2023	59.7 km ²	−1.2 km ²	−1.97%
2024	57.8 km ²	−1.9 km ²	−3.18%
2025	55.7 km ²	−2.1 km ²	−3.63%

Spatial change

Compare masks from two dates to identify pixels/regions where detected glacier extent differs. This creates a spatial change layer that can be displayed on a map.

Historical significance

A later stage can compare the latest change against the glacier's own historical behavior. If most year-to-year changes are small but one interval is substantially larger, the system can label it as **notable** under an explicitly documented criterion.

8. DASHBOARD & USER EXPERIENCE

The dashboard is where the AI and geospatial analytics become useful to researchers and other users.

GLACIER MONITOR	G001 • Selected Glacier
Current ice area 55.7 km ²	Observed change −3.63%
Historical trend 2020 → 2025 62.4 → 55.7 km ²	Status NOTABLE CHANGE Requires investigation
Area history 	Spatial change map [Glacier map / change layer]

Suggested user flow

1. User selects a glacier.
2. System shows latest satellite observation and AI-derived ice mask.
3. System displays current ice-covered area and comparison with previous observations.
4. User opens the historical chart to inspect the time series.
5. User opens the change map to see where spatial difference occurred.
6. If the change meets the documented criterion, the dashboard highlights it for investigation.
7. User can inspect source observations and generate a report.

9. TECHNICAL SYSTEM ARCHITECTURE

Frontend HTML / CSS / JavaScript Charts + interactive map	Django / REST API Authentication Glacier + observation APIs Analysis jobs	AI + Geospatial Engine Segmentation Mask processing Area + change analysis
PostgreSQL Glaciers Observations Measurements Analysis results	Image / Object Storage Raw imagery Processed rasters Masks Reports	Security Layer Authentication Authorization Validation Logging / monitoring

Core data flow

Satellite image	→	Preprocessing	→	Segmentation model
Segmentation model	→	Ice mask	→	Geospatial area
Geospatial area	→	Historical DB	→	Change analytics
Change analytics	→	Django API	→	Dashboard / reports

10. TEAM IMPLEMENTATION PLAN

AI MEMBER 1	AI MEMBER 2	CYBER MEMBER 1	CYBER MEMBER 2
Satellite preprocessing Segmentation dataset Mask generation Model training	Area calculation Historical measurements Change detection Notable-change analysis Optional forecasting	Authentication Authorization API security Upload validation Input validation	Secure deployment Logging & monitoring Database security Integrity controls Security testing

Interface between AI members

AI Member 1 produces a georeferenced image + predicted ice mask. AI Member 2 consumes that mask together with raster metadata and produces measurements and analytics.

Example result: glacier_id = G001 • date = 2025-01-15 • ice_area_km2 = 55.7 • change_km2 = -2.1 • change_percent = -3.63 • status = NOTABLE

11. DEVELOPMENT ROADMAP — WHAT WE BUILD FIRST

Stage	Build	Purpose
PHASE 1	1 glacier + multiple satellite dates	Understand data and prove the end-to-end concept.
PHASE 2	Preprocessing + labeled masks	Create the AI-ready dataset.
PHASE 3	Segmentation model	Predict glacier-ice pixels.
PHASE 4	Area measurement	Convert masks into geospatial ice-covered area.
PHASE 5	Historical database	Store observations and measurements.
PHASE 6	Change analysis	Compute area and spatial changes.
PHASE 7	Dashboard	Make results interactive and traceable.
PHASE 8	2–3 glaciers	Prove the system generalizes beyond one glacier.
PHASE 9	Advanced monitoring	Automated ingestion, anomaly methods, climate data, forecasting where scientifically justified.

12. CURRENT NEXT STEP

We are currently at Phase 1 — Data Foundation.

- Choose one real glacier.
- Learn its geographic location and basic characteristics.
- Select an appropriate satellite dataset.
- Collect multiple good observations of the same glacier over time.
- Inspect one observation carefully before collecting the full set.

Only after the data is understood and organized should we move into mask creation, model training, area calculation and the monitoring dashboard.

FINAL PROJECT MENTAL MODEL

Observe → Identify ice → Measure → Store → Compare → Detect spatial change → Assess notable change → Visualize → Investigate